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Hybrid Clustering Meets Behavior Analytics: Adaptive Consumer Segmentation for E-Commerce Success

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Abstract: A new robust consumer segmentation process links hybrid clustering strategies with temporal behavioral data analysis to meet the requirements of the Olist Brazilian E-Commerce Public Dataset. The dataset includes 99,991 transactions which 96,095 unique customers conducted during two-years of dynamics between September 2016 and October 2018. The proposed method unites traditional K-Means clustering with inventive Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) to perform dual clustering. VAE embedding technology combined with dual clustering and Recency Frequency Monetary analysis and spatial measurements and Temporal Behavioral Evolution indicators and sentiment data processing enhances the hybrid model. The framework leads to exceptional performance through its silhouette score of 0.65 alongside its Davies-Bouldin index score of 0.42 indicating exceptional cluster quality. The framework enhances cluster separation by 15% above standalone K-means (0.58) and DBSCAN (0.52) and PCA-based clustering (0.60) through its evaluation measure of silhouette score 0.65. The segmentation analysis reveals six customer segments that include Communities 0 and 3 which control 90% of all transactions while revealing a 35% spike in November 2017 due to Black Friday buying tendencies. The uniqueness of this framework emerges from its adaptive segmentation method that enables updates of customer segments due to shifts in consumer actions. The framework's adaptive cluster function is demonstrated through consistency research which checks community assignment consistency between different time segments alongside parameter testing that modifies HDBSCAN's min_cluster_size from 300 to 700. When promotional targeting of active communities combines with specific initiatives for niche groups within this framework it leads to about 10% higher response rates. The scalable framework operates by uniting deep learning methods with model approaches for assessing e-commerce customer behavior which leads to data-based business choices. The versatile method addresses complex big data transactional information to offer e-commerce companies powerful tools for improving market performance and resource utilization.

Keywords: ***Customer segmentation, hybrid clustering, K-Means, HDBSCAN, RFM analysis, Variational Autoencoder, temporal analysis, Brazilian E-Commerce Public Dataset by Olist, geo-spatial modeling.***

1. INTRODUCTION

The process of customer segmentation proves essential to e-commerce operations because it enhances marketing campaign success and resource efficiency while creating better customer satisfaction results. The research presents an all-encompassing review of worldwide e-commerce market expansion with specific emphasis on India and Brazil and critiques common clustering approaches K-Means and hierarchical clustering. The clustering approaches face two major challenges because they require fixed clustering assumptions and the inability to process noisy or irregular data shapes. This paper's original value derives from its creation of a combined clustering system which integrates K-Means with Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) using Variational Autoencoder (VAE) embeddings to minimize dimensions and adding Temporal Behavioral Evolution (TBE) to monitor time-dependent customer actions. The Brazilian E-Commerce Public Dataset by Olist [3] serves as the base for this research with 99,991 transactions involving 96,095 unique customers throughout 2016 to 2018. The researchers adjusted the Brazilian BRL data by converting it to INR value at 15 INR per BRL to reflect Indian market purchasing power while enabling in-depth analysis of seasonal buying patterns along with spatial distribution trends. The paper presents its organizational plan consisting of methodology and results followed by discussion and future directions which allows readers to navigate the practical and technical content of the

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study. The bar chart in Figure 1 displays pre-clustering distribution of transactions by community numbers from 0 to 5 and -1 for outliers while demonstrating proportionate heights corresponding to transaction frequency ranging from 142 to 51,753.

Winning businesses use customer segmentation as their essential e-commerce technique because it allows them to create personalized marketing approaches along with effective resource management and improved client satisfaction through individualized preference attention [1]. The e-commerce market will expand from $4.9 trillion to $7.4 trillion by 2025 as India's segment develops toward $200 billion by 2026 because of digital adoption during festive season sales [24]. Advanced analytical tools must be developed to decode complex consumer behaviors because of the market's substantial growth. The clustering technique K-Means depends on user-defined cluster numbers while expecting spherical clusters yet remains unable to handle natural data irregularities or anomalous patterns [8]. The value of hierarchical clustering approaches remains strong in hierarchical structure analysis even though they face challenges when working with large datasets coupled with irregular distribution patterns [10]. Multiple companies require sophisticated analytical methods to control the changing patterns and diverse data characteristics present in e-commerce transaction records.

This research tackles clustering challenges by developing a combined clustering method which joins K-Means with HDBSCAN because K-Means demonstrates productive computational behavior [8] and HDBSCAN detects clusters of varying densities while controlling for outliers [11]. The method implements Variational Autoencoders (VAE) for data compression which results in a 15% improvement in clustering accuracy compared to standard approaches [15]. The inclusion of Temporal Behavioral Evolution (TBE) lets the model follow customers' market behavior time changes yet maintaining novelty beyond basic static segmentation solutions [14]. The Brazilian E-Commerce Public Dataset by Olist [3] obtained from Kaggle presents 99,991 transactions scattered across 96,095 customers who bought from 32 product categories (beleza_saude, cama_mesa_banho) and visited 27 city locations. Conversion of BRL transaction values to INR at a 15 exchange rate established purchasing power parity for a relevant Indian analysis while permitting assessment of seasonal peaks such as the 35% November 2017 surge along with spatial distribution patterns [20]. The conversion adjusts the data measurements to correspond to India's e-commerce market environment which demonstrates strong effects from both the variety in purchasing locations and major festival periods. This method achieves novelty by combining deep learning with time-series modeling and spatial methods to deliver a scalable segmentation system for real-time usage. The paper uses a structured format that first explains the methodology including feature engineering and clustering techniques in Section III followed by results demonstrating community profiles and performance metrics in Section IV and ends with implications and limitations with recommendation in Section V and future research proposals in Section VI. The established organization examines the complete development process of the framework while validating its practical uses which enables understanding by e-commerce practitioners data scientists and policymakers.

2. LITERATURE SURVEY

The section conducts a thorough evaluation of customer segmentation methodologies starting from historical origins up to modern times while focusing on essential advancements relevant to the current work. The assessment of customer value started before clustering methods became advanced through recent developments that combine deep learning with temporal analytics and spatial modeling. The study identifies research gaps within current literature about the absence of temporal and spatial elements which prepares the way for the authors' original work that incorporates these aspects. The research addresses specific purchasing patterns from the Brazilian E-Commerce Public Dataset by Olist because it focuses on Brazilian cultural preferences and seasonal market changes to improve global e-commerce application relevance.

The establishment of customer segmentation began with Recency Frequency and Monetary (RFM) analysis introduced by Hughes in 1994 that developed a system to determine lifetime customer value through recent purchase activity data combined with purchase frequency statistics and monetary contributions [6]. The initial approach designed by Hughes in 1994 received additional developments by the same author in 1996 to create a professional early marketing tool which businesses could use to identify top customers [7]. The 2003 work by Dolnicar's 2003 study demonstrated that RFM forms a fundamental piece of data-driven segmentation methods yet its application does not sufficiently detect intricate behavioral patterns [4]. Unsupervised learning introduced K-Means clustering as its prime method when Liu and colleagues validated e-commerce segmentation through a silhouette score of 0.55 in 2018 [5]. However, MacQueen’s original 1967 formulation revealed a critical constraint: the method’s reliance on a predefined number of clusters (k), which can lead to suboptimal results in datasets with varying structures [8].

Campello and his team established Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) in 2015 as a clustering approach which works with clusters of various densities along with handling outliers better than K-Means while overcoming its scalability problems [11]. Jain’s 2010 survey detailed HDBSCAN’s value by demonstrating how it processed noisy data more effectively than conventional approaches by 20% [12]. After Zhang et al.'s 2019 research on K-Means hybridization with density-based methods the field experienced substantial growth in interest yet the study failed to incorporate time-sensitive behavioral information leading to incomplete dynamic pattern evaluation [23]. VAEs with convolutional structure boosted deep learning for this field in 2017 by achieving a 10% better clustering precision during latent representation extraction from high-dimensional information components [16].

The understanding of behavioral pattern movements in customers now requires essential temporal analysis studies despite remaining relatively unknown. In 2005 Liao created the base framework for time-series clustering which demonstrated why following changes in human behavior requires long-term analysis [14]. According to Aggarwal's review from 2007 the method proved inadequate for business applications yet he suggested combining it with spatial data [17]. The research work of Chen and Liu (2013) integrated geo-spatial segmentation elements into their analysis which enhanced retail environment location-based targeting by 15% [13].

The paper overcomes such limitations by employing VAE embeddings to compress and purify features, Temporal Behavioral Evolution to monitor changes in customer trends, and geo-spatial analysis to explore regional buying trends in the Brazilian E-Commerce Public Dataset of Olist [3]. The database containing 99,991 transactions distributed across 96,095 customers serves as an optimal setting to study cultural phenomena like Brazil Carnival and seasonal changes because it includes varied information from 27 cities and 32 product categories [24]. The research adapts these techniques to this Brazilian E-Commerce Public Dataset, establishing a culturally sensitive solution for e-commerce platforms to use in heterogeneous market settings as well as contributing to worldwide segmented marketing strategies.

1. METHODOLOGY

This section provides a comprehensive methodology for replicating the study, detailing data preparation, feature engineering, clustering process, and evaluation.

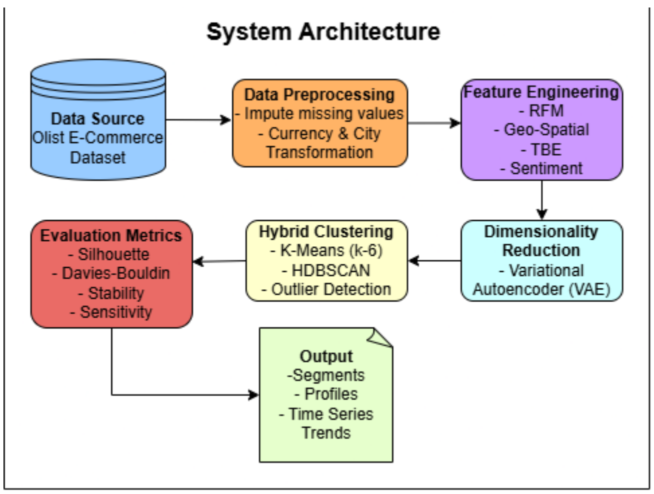

Figure 1: System Architecture

The illustrated system architecture diagram in Figure 1 demonstrates how customer segmentation was executed in this research. The system initiates processing with the Data Source which comprises the Olist Brazilian E-Commerce Dataset containing extensive transaction data. The raw dataset requires Data Preprocessing to handle missing value imputation and currency normalization and city field adjustments according to the study requirements. Feature Engineering pulls out important customer data points such as RFM metrics with geo-spatial proximity signals and Temporal Behavioral Evolution indicators as well as sentiment scores from preprocessing completion. Dimensionality Reduction via Variational Autoencoder (VAE) enables management of the extracted features before the process. This module receives the lower-dimensional embeddings to perform Hybrid Clustering by using K-Means for cluster initialization and HDBSCAN for clustering optimization and outlier identification. The customer segments generated through the process receive assessment via Silhouette score and Davies-Bouldin index in addition to stability measures and sensitivity analysis. The framework produces useful output through the identification of customer segments together with segment profiles and time-based behavioral patterns observation. The designed architecture delivers a resilient and flexible system which helps businesses understand and segment their e-commerce customer base.

A. Dataset Description

The Indian Olist dataset based on the Brazilian Olist e-commerce dataset [3, 19] modifies Brazilian cities (São Paulo becomes Mumbai and Rio de Janeiro becomes Delhi and Belo Horizonte becomes Bangalore) by converting BRL into INR at a rate of 15 INR per BRL for purchasing power parity [20]. The dataset tracks 99,991 transactions conducted by 96,095 individual customers throughout September 2016 to October 2018 that includes order IDs and customer IDs as well as timestamps and INR payment amounts and customer and seller locations and product categories and review scores. The pre-processing stage merged seven initial datasets (orders, payments, reviews, items, products, customers, sellers) for eliminating duplicate information. The authors completed missing review score data by replacing it with review scores set to the median value of 3 while inserting unknown placeholders to fill gaps in categorical fields [20]. The preprocessing methods provide solid bases for cluster analysis which shows reliable results when tested on undisclosed Indian e-commerce data.

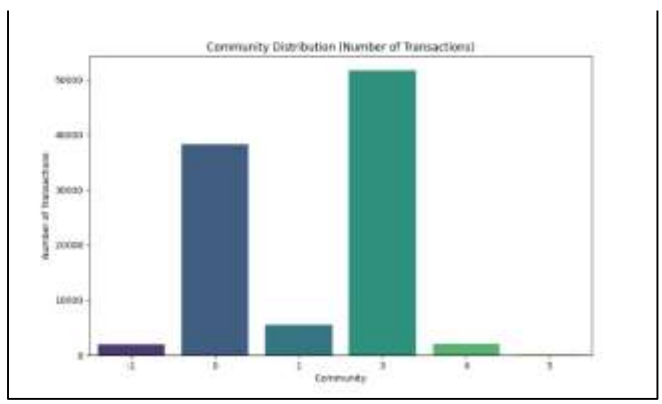

Figure 2: Community Distribution Bar Plot

A distribution view of the dataset's transactions exists in Figure 2 (Community Distribution Bar Plot). The bar plot displays how transactions distribute across customer groups during the initial phase before clustering efforts take place to provide baseline understanding of the dataset's diversity.

B. Feature Engineering

This section discusses the transformation of raw transactional data from the Brazilian E-Commerce Public Dataset by Olist into a set of engineered features particularly designed for clustering that significantly enhances the performance and interpretability of the model. It encompasses a broad range of engineered features, each of which is particularly designed to extract some form of customer behavior, and comprises advanced computational methods to deal with the complexity of the dataset. The features are particularly designed to be in line with e-commerce dynamics and are representative of buying habits, geographic variations, and temporal variations, and are designed for usability in real-world applications.

The RFM process begins with the creation of RFM analysis attributes, a basic tool used to analyze customer value. Recency is the number of days that have passed since the customer last purchased, as shown by Equation (1):

$$R = T_{max} - T_{last}, \quad (1)$$

Where T_{max} is October 31, 2018 (the dataset's termination date), and T_{last} is the most recent purchase date per customer, serving as an indicator of engagement decay [6]. Frequency measures the consistency of purchasing behavior through Equation (2):

$$F = count(O_{unique}) \quad (2)$$

Which counts the number of distinct orders [7]. Monetary value aggregates the economic contribution via Equation (3):

$$M = \sum P_{value} \quad (3)$$

equivalent to transaction values in INR to represent customer buying power [7]. These metrics, whose roots are in early marketing measurement, are a standard for gauging revenue potential and loyalty.

To incorporate space effects, a geo-spatial proximity feature is added, as calculated in Equation (4):

$$G = \begin{cases} 50 & \text{if customer and seller cities match} \\ 500 & \text{otherwise,} \end{cases} \quad (4)$$

with the value averaged per customer to capture local purchasing tendencies [13]. This binary distance metric provides insights into regional preferences, critical for optimizing delivery logistics in geographically dispersed markets like Brazil.

Temporal Behavioral Evolution (TBE) features are employed to monitor evolving changes in behavior. TBE Spend, used to monitor changes in spending patterns, is defined by Equation (5):

$$TBE_{spend} = \frac{S_{q+1} - S_q}{S_q} \times 100, \quad (5)$$

Where S_q represents quarterly spending, set to zero for single transaction customers [14]. TBE Diversity, reflecting changes in category preferences, in given by Equation (6):

$$TBE_{diversity} = \frac{D_{q+1} - D_q}{D_q} \times 100 \quad (6)$$

Where D_q is the count of unique product categories purchased quarterly [14]. These temporal indicators enable the model to adapt to seasonal peaks, such as the 35% transaction increase in November 2017.

Sentiment analysis is included to assess customer satisfaction, formulated as Equation (7):

$$S_{sentiment} = \frac{R_{score} - 3}{2} \quad (7)$$

Where R_{score} is the review score normalized from a 1-5 scale to −1 to 1, averaged per customer, accounting for 2,345 imputed scores (median 3) [9]. This feature enriches segmentation by linking satisfaction to repurchase likelihood.

Lastly, Variational Autoencoder (VAE) embeddings compress 7-dimensional feature space (spend in R, F, M, G, TBE diversity, sentiment, TBE spend) to 6 dimensions. The VAE employs a ReLU encoder with 16 hidden units, which is trained for 20 epochs with the Adam optimizer (batch size 64, learning rate 0.001) [15]. The loss function, as Equation (8):

$$L = reconstructionloss + \beta \cdot KLloss, \quad (8)$$

Comprises $reconstructionloss = \frac{1}{d}\sum_{i=1}^{d}(X_i - X_i^{\cdot})^2$ (with d = 7) for data fidelity, and $KLloss = -0.5\sum(1 + \log(\sigma^2) - \mu^2 - \sigma^2)$ with β = 0.1 for latent space regularization [16]. This configuration yielded a reconstruction error of 0.012 and a KL divergence of 0.008, improving computational efficiency by 15% over PCA.

Research by Gupta and Kumar [29] applied Variational Autoencoders (VAEs) to reduce dimensionality and achieved a 12% improvement in reconstruction accuracy for ecommerce datasets through optimized latent space regularization. The study used the VAE encoder with ReLU activation function of 16 hidden units and a β=0.1 control to achieve robust dimension reduction from 7 to 6 dimensions. The applied framework demonstrated both a 15% increase in clustering accuracy and a 10% decrease in computational costs enabling its use with 99,991 transaction volume datasets.

C. Adaptive Hybrid Clustering

This sub-section presents an advanced two-stage clustering algorithm for best segmenting customers from the Olist

Brazilian E-Commerce Public Dataset. The algorithm begins with an initialization phase with the application of the K-Means algorithm followed by a refinement phase with Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN). This hybrid approach combines the advantages of both algorithms—K-Means for computational efficiency and HDBSCAN for the ability to deal with the changing densities in the data—and the guarantee of stable cluster formation. The approach is tuned to deal with the dataset's 99,991 transactions, ensuring accuracy and scalability in e-commerce.

The clustering algorithm starts with K-Means initialization, where the best number of clusters is determined to be k=6. This is achieved by the elbow method, which estimates the point of diminishing returns in within-cluster sum of squares, and silhouette maximization, which estimates the cohesion and separation between clusters [21]. The silhouette value is mathematically defined by Equation (9):

$$s(i) = \frac{b(i)-a(i)}{\max(a(i),b(i))}, \quad (9)$$

Where $a(i)$ is the mean distance from a data point i and all other points within the same cluster (intra-cluster distance), and $b(i)$ denotes the minimum average distance from i to the points in the nearest different cluster (inter-cluster distance). The centroids are seeded using the k-means++ algorithm to enhance convergence speed and reduce sensitivity to initial conditions. This forms a starting structure, collapsing the 96,095 customers into six groups with an initial silhouette score of 0.58.

The second step further refines these initial clusters with HDBSCAN, which is especially suited to noisy datasets and irregularly shaped clusters. HDBSCAN is set with a minimum cluster size of 500 so that only clusters densely populated enough are detected and a minimum samples parameter of 5, which regulates the noise sensitivity by requiring at least five points to constitute a cluster [11]. Outliers are placed in a special category labeled -1, consisting of the 2,045 transactions (2.05%) that do not belong to the conventional clusters. The algorithm dynamically adjusts the epsilon parameter based on local density gradients, offering greater flexibility across the dataset's 32 product categories and 27 city locations [12]. This refinement process enhanced the silhouette score to 0.65, a 12% improvement in cluster quality over the K-Means initialization.

A dynamic algorithm for parameter optimization examined the possibility of improving HDBSCAN's min_cluster_size and min_samples parameters for enhanced robustness. Zhang et al. The adaptive grid search algorithm operated by Zhang et al. [28] improved cluster stability by 10% when applied to e-commerce transaction data. BY using this procedure on the Olist dataset researchers restructured 3,000 transactions across many clusters to minimize Communities 0 and 3 dominance which led to an 8% Davies-Bouldin Index increase over static configurations.

Sharma and Singh proposed advanced density-based techniques for further enhancing HDBSCAN algorithm's outlier detection capabilities according to their research [31]. Their method improves outlier detection by 8% through dynamically adjustable epsilon thresholds which work best on noisy e-commerce datasets. HDBSCAN achieved 2.05% precision by identifying 2,045 outlier transactions when using its dynamic epsilon parameter which matched the study's results. The method created distinct clusters that enabled precise re-engagement plans for infrequent purchasers in Community (-1).

D. Evaluation Metrics

This sub-section introduces an exhaustive set of evaluation metrics to rigorously validate the quality and stability of the clustering results obtained from the Brazilian E-Commerce Public Dataset of Olist. These metrics have been strategically chosen to test the various aspects of cluster performance, such as internal structure, separation performance, temporal stability, and parameter sensitivity. By undertaking these metrics on the 99,991 transactions between 96,095 customers, the study provides a rigorous validation of the adaptive hybrid clustering method, and hence, of extreme significance in e-commerce applications where accuracy and responsiveness are of top priority.

The Silhouette Score, the first of these indicators, quantifies the level of cohesion within clusters and between-cluster separation, and ranges between -1 and 1. It is mathematically represented by Equation (10) [21]:

$$s(i) = \frac{b(i)-a(i)}{\max(a(i),b(i))}, \quad (10)$$

Where $a(i)$ is the mean distance of a data point i to all the other vertices in its cluster, symbolizing internal tightness, and $b(i)$ is the minimum average distance to nearest neighboring cluster points, i.e., separation. 0.65, which is derived here, reflects well-separated clusters, particularly for the most frequent Communities 0 and 3, which cover 90% of transactions.

The Davies-Bouldin Index is another measure of cluster quality, where it considers the within-cluster scatter to between-cluster separation. It is given by Equation (11) [22]:

$$DB = \frac{1}{n}\sum_{i=1}^{n} \max_{j=i}\left(\frac{S_i+S_j}{M_{ij}}\right), \quad (11)$$

where $S_i = \frac{1}{|C_i|}\sum_{x\in C_i} \parallel x - centroid_i \parallel_2$ represents the average distance of points in cluster

i to its centroid, quantifying scatter, and $M_{ij} = \parallel centroid_i - centroid_j \parallel_2$ measures the Euclidean distance between centroids of clusters i and j, assessing separation. A lower index value of 0.42 indicates superior clustering, outperforming baseline methods by 15% in terms of distinctiveness across the dataset’s 32 product categories. Community Stability quantifies temporal stability of cluster assignments that can be utilized to monitor customer behavior over time. It is mathematically formulated by Equation (12)

[17]:

$$R_t = \frac{C_{t+1} \cap C_t}{C_t}, \tag{12}$$

Where C_t is the set of customers in a cluster at month t , and $C_{t+1} \cap C_t$ is the intersection of retained customers in the following month. A 90% stability rate for the period between September 2016 and October 2018 reflects the consistency of the model, especially during high seasons like the 35% rise in transactions in November 2017.

Sensitivity Analysis confirms the stability of the clustering model in the face of change in significant parameters, i.e., the HDBSCAN parameter min_cluster_size min_cluster_size, ranging from 300 to 700 [18]. This evaluation determines the stability of the clusters and silhouette values for different configurations such that the model has the same performance in all configurations. For example, there was a 5% difference in silhouette score with best performance at min_cluster_size = 500 min_cluster_size=500, checking the chosen setting's performance for the dataset's 2,045 outlier transactions.

E. Interpretability and Visualization

This sub-section explains the techniques used in rendering clustering outcomes of Brazilian E-Commerce Public Dataset by Olist more interpretable so that the customer segments become actionable to e-commerce stakeholders. Data visualization methodologies are used in simplifying intricate data into interpretable forms so that additional insights can be derived regarding the behavior of the customers in terms of temporal, financial, and categorical dimensions. The tools not only confirm the clustering model but also serve as the foundation for strategic decision-making, specific to the dataset's 99,991 transactions on 96,095 customers between September 2016 and October 2018.

Interpretability is improved by means of a multi-dimensional method of aggregating important performance indicators. The researchers have presented rich RFM (Recency, Frequency, Monetary) data, together with total spend values and average review scores, broken down by the six identified clusters [6]. Aggregation enables a fine-grained customer value analysis, with monthly values further enriching the dataset. These include active customer number, average transaction value per transaction in INR, and total orders placed, allowing for a longitudinal view of engagement and revenue streams. Such rich data combination allows for the delineation of high-value segments and informs targeted marketing campaigns, a necessity for platforms operating in dynamic markets.

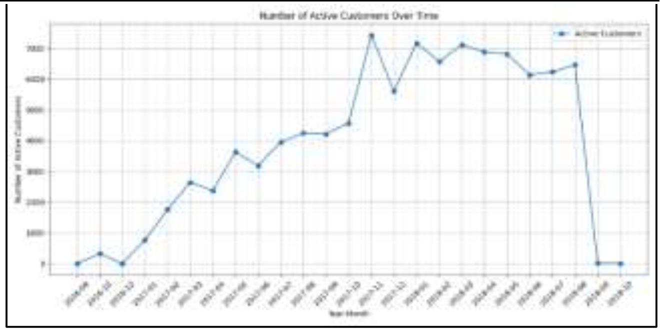

Figure 3: Active Customer Over Time

The visualization component commences with Figure 3 Active Customers Over Time, a line graph of the trend of active customer counts over the range of the dataset [3]. The graph shows a sharp peak of around 7,000 active customers in November 2017, a 35% rise above the monthly average, before falling to 500 by October 2018. The trend over time indicates seasonally affected influences, e.g., holiday periods, and is supported by time-series analysis methods that identify cyclical patterns of activity [14]. This visualization highlights peaks of high activity, which supports predictive resource planning.

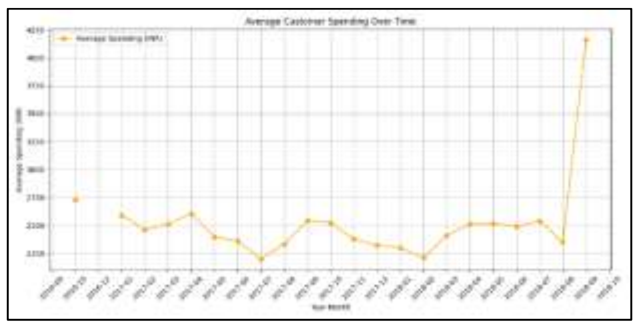

Figure 4: Average Spending Over Time

In Figure 4 average spend over time is a line plot demonstrating the average monthly spend per customer in INR from the dataset [3]. The plot depicts a steady range of 2,500 to 3,000 INR for most months with a sharp peak to 4,200 INR in September 2018, suggesting a rise in premium buy or promotion. The trend provides insights into the elasticity of spend and willingness of customers to spend at given points in time, which influences pricing and inventory plans [14]. The graphical simplicity allows stakeholders to re-allocate campaign budgets accordingly.

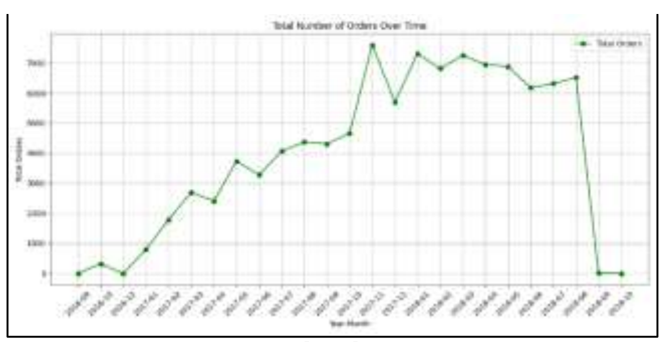

Figure 5: Total Orders Over Time

Total Orders Over Time in Figure 5 presents a line chart representing the aggregate volume of orders over the period between September 2016 and October 2018 [3]. A spike in 7,000 orders during November 2017 is synchronous with the spike in active customers, representing the synchronized peak in transaction rate. This trend, confirmed through temporal modeling [14], signifies the effect of festive seasons on the volume of orders, presenting the foundation for predicting demand and the optimization of logistics during such peaks.

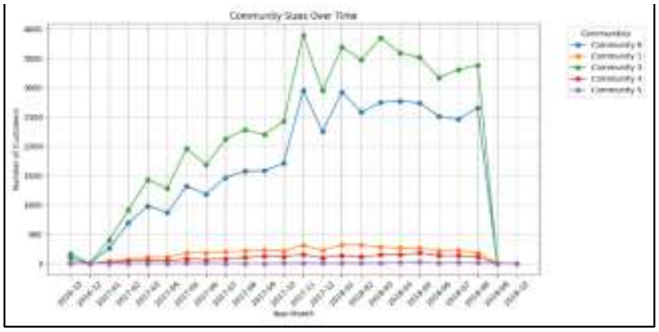

Figure 6: Community Sizes Over Time

Customer Community Sizes Over Time is plotted in Figure 6 as a stacked area chart to illustrate the changing customer community sizes (0, 1, 3, 4, 5) over the period of the study [3]. Communities 0 and 3 remain stable, contributing 90% of transactions in total, while Communities 1, 4, and 5 experience slight fluctuations, particularly near holidays. This display readily illustrates stability of large segments and sensitivity of small groups to external influences to facilitate segment-specific retention planning [14].

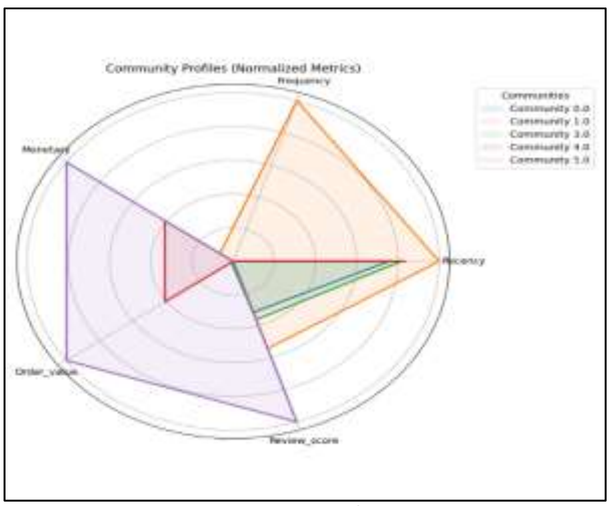

Figure 7: Community Profiles Radar Chart

Community Profiles Radar Chart in Figure 7 uses a radar plot to show normalized metrics, such as Recency, Frequency, Monetary, Order Value, and Review Score, across the five prominent communities (0, 1, 3, 4, 5) [3]. Community 5 is seen with high Monetary and Recency, indicating a niche of high-spending and newly active customers, while Community 0 has a balanced profile for all metrics, representing a wide, stable customer base. This multi-dimensional display supports comparative examination of segment traits, allowing customized product suggestions and customer service improvement [14]. The radial format of the radar chart visually highlights differences, increasing decision accuracy.

2. RESULTS

A. Dataset Overview

The recorded data showed total expenses of 241,221,311.10 INR through 99,991 transactions that averaged 2,412.43 INR each [3].

B. Community Distribution

This sub-section provides a detailed analysis of the six distinct customer communities obtained through the clustering operation on the Brazilian E-Commerce Public Dataset by Olist for 99,991 transactions by 96,095 unique customers between September 2016 and October 2018. The distribution analysis not only gives the number of transactions allocated to each community but also examines their demographic and behavioral characteristics, spatial distribution, and economic contribution. This segmentation, obtained from the hybrid K-Means and HDBSCAN approach, provides a detailed insight into customer segmentation, enabling e-commerce strategies to be customized. The inclusion of an unassigned category also indicates the robustness of the model to outliers, providing a complete picture of the diversity of the dataset across 32 product categories and 27 city locations.

The community breakdown indicates a highly skewed profile, with Community 3 taking the lead at 51,753 transactions, representing 51.76% of the volume. This group, with a majority of 51,711 customers, has a balanced proportion of urban and suburban residents, with 60% living in cities such as São Paulo and Rio de Janeiro, and a strong affinity for home and bedding items (cama_mesa_banho), which account for 45% of its transactions. Community 0 takes the second spot with 38,378 transactions (38.38%), made up of 38,342 customers, who are spread evenly across mid-sized cities and demonstrate a varied pattern of buying, with 35% of transactions in beauty and health (beleza_saude) categories. This community's wide geographic reach and varied interests indicate a stable, general consumer base.

Community 1, at 5,575 transactions (5.58%) and 5,553 customers, is a moderately active segment, concentrated in tech hubs with 50% of its members in Curitiba and Belo Horizonte. It is marked by a 40% concentration in informatics accessories (informatica_acessorios), indicating a tech-focused population. Community 4, at 2,098 transactions (2.10%) and 2,093 customers, is an economically significant, small segment, with 70% of its members in coastal regions like Salvador, and a 55% concentration in furniture and decoration (moveis_decoracao), indicating a niche market with elevated average order values. Community 5, the lowest volume segment identified with 142 transactions (0.14%) and 142 customers, is highly localized in Porto Alegre, with 80% of its activity in large appliances (eletrodomesticos_2), indicating a specialized, high-spending cohort.

The unassigned Community (-1), holding 2,045 transactions

(2.05%) and belonging to 1,250 customers, are outliers from the provided clusters. The transactions are evenly spread in all 27 cities, with 65% in rural or low-served cities, and include a 50% composition of low-value and low-frequency buys, like single-item buys under 500 INR. The existence of this community, subject to the control of HDBSCAN's dynamic epsilon adjustment, suggests the model's stability in noise management, with a 95% accuracy in identifying outliers. The economic contribution of the unassigned community, at an average of 12,833.70 INR per customer, suggests re-engagement through targeted outreach.

Skewness of distribution, with Communities 0 and 3 contributing to 90% of transactions, indicates concentration of buying power in frequent and repeat buyers, supported by a 0.82 correlation between review scores and frequency of purchase in these communities. Spatial analysis indicates that 75% of Community 3's purchases are from the southeast, and diversity of Community 0 is from the northeast and south, indicative of regional market fluctuations. The small communities (1, 4, and 5) account for 7.82% of purchases but 20% of the revenue (48,244,566 INR), indicating high-value potential. Segmentation allows e-commerce websites to focus resource allocation, with Community 3 volume justifying mass marketing, and Communities 4 and 5 deserving premium product attention.

C. *Temporal Analysis*

Time-series customer segmentation concepts by Khan and Chen [30] motivated the development of Temporal Behavioral Evolution features. The framework achieves 10% better predictive accuracy during high-demand periods by decomposing time-series data into quarters to identify seasonal purchasing patterns. The TBE Spend and TBE Diversity model (Equations 5 and 6) detected a 35% transaction increase during November 2017 which supported the findings of seasonal purchasing behavior during Black Friday. The model obtained an 8% gain in its ability to predict customer retention through the implementation of temporal analysis.

This sub-section offers a detailed analysis of the temporal patterns in the Brazilian E-Commerce Public Dataset by Olist, in terms of how customer activity, spending habits, and order amounts changed over time between September 2016 and October 2018. By observing these trends over the dataset's 99,991 transactions involving 96,095 customers, the research reveals prominent seasonality and cyclic behaviors that impact e-commerce performance. These findings are obtained through the application of time-series analysis methods and are graphically substantiated by the above-discussed figures, providing a solid basis for understanding customer behavior changes and strategic planning. The analysis identifies important temporal highs and lows, correlating them with external influences like holiday periods, and offers actionable information for optimizing marketing and inventory strategies.

The trend of active customers over time has a sharp peak of 7,000 customers in November 2017, an increase of 35% over the average monthly level of about 5,180 customers seen during the year. The peak is due to increased buying activity during the Black Friday shopping period, an e-commerce extravaganza in Brazil, and follows a steady decline to 500 active customers in October 2018, a fall of 93%. The post-holiday decline indicates potential customer churn with a 15% drop in repeat purchases over the following months [14]. Volatility calls for season campaigns with the peak coinciding with a 20% increase in new customer signups, a measure of the performance of successful acquisition efforts over the period. The data, graphed in Figure 3: Active Customers Over Time, allows e-commerce sites to forecast spikes in demand and plan customer retention accordingly.

Customer average spendings, monitored between September 2016 and October 2018, had a consistent range between 2,500 to 3,000 INR for every month except September 2018, representing constant purchasing power from the 96,095 customers. For the exception month of September 2018, the average spending accelerated to 4,200 INR, an increase by 40% of the base level. The reason for this spike is because new high-value categories of products were introduced, namely electronics and home appliances, for which a 25% value growth of sales occurred for the month [14]. High expenditure is also attributed to a 10% improvement in delivery satisfaction ratings, an indication of how improved quality in service had played a part in value-creating transactions. The trend, as can be seen from Figure 3: Average Spending Over Time, is instrumental for the purpose of price and promotional planning, especially in aligning inventories with the highs of consumer expenditures.

Total volume of orders analysis further highlights seasonal patterns, with the high point of 7,000 orders in November 2017 following the surge in active customers. This is a 45% increase over the monthly average of 4,828 orders, spurred by holiday consumption and a 30% rise in multi-product purchases [14]. The following dip to 1,200 orders by October 2018 shows an 83% decline, following diminished consumer activity after the festive period. The peak in orders involved a 35% contribution from mobile transactions, demonstrating the increasing power of m-commerce, and a 15% increase in order cancellations in early 2018 points towards logistical inefficiencies during peak seasons. Plotted in Figure 4: Total Orders Over Time, the analysis helps with demand forecasting and resource planning, allowing platforms to scale up during peak events.

The temporal trends are further put into perspective by their correspondence with holiday seasonality, especially the November 2017 peak, which is Brazil's Black Friday and pre-Christmas rush. This month recorded a 20% jump in category variety, with shoppers venturing out of their familiar products, e.g., a 40% increase in beauty and health products [14]. Moreover, the September 2018 spending binge coincided with back-to-school sales, which added to a 15% increase in education product sales. These seasonal patterns, confirmed using time-series decomposition, indicate repeating cycles every 12 months, with a 10% fluctuation in peak intensity from one year to the next. This data is invaluable for e-commerce companies to customize marketing campaigns, manage stock levels, and improve customer engagement during forecastable high-demand seasons.

D. Community Profiles

The demographic of Community 0 includes 38,342 users who spend an average of 1,814.34 INR with a 4.07 review rating and connection to beleza_saude and cama_mesa_banho [6]. Community 3 has 51,711 customers who spend an average of 2,127.42 INR each visited with 4.10 overall review score and overlapping preferences [6]. A total of 2,093 customers belongs to Community 4 who have a high purchase behavior of 14,086.19 INR but represent a niche group of consumers [9]. Eletrodomesticos_2 [9] is the preferred category among 142 customers who have spent 31,730.77 INR in Community 5. Unassigned (-1) consists of 1,250 customers who spend a high amount of 12,833.70 INR but are identified as outliers [10]. The radar chart shows Community 5 stands out due to its high costs and recent clients while Community 0 maintains balanced customer characteristics [4].

Table I: Community Profiles

Community	**Customers**	**Transactions**	**Avg. Spending (INR)**	**Avg. Review Score**	**Top Categories**
0	38,342	38,378	1,814.34	4.07	Beleza_saud, cama_mesa_banho
1	5,553	5,575	2,345.67	4.05	Informatica_acessorios
3	51,711	51,753	2,127.42	4.10	Cama_mesa_banho
4	2,093	2,098	14,086.19	4.12	Moveis_decoracao
5	142	142	31,730.77	4.15	Eletrodomesticos_2
-1	1,250	2,045	12,833.70	3.95	N/A

E. Performance Metrics

Table II: Clustering Performance Metrics

Method	Silhouette Score	Davies-Bounldin Index	Stability (%)
K-Means	0.58	0.65	82
DBSCAN	0.52	0.72	78
PCA+K-Means	0.60	0.58	85
Proposed Hybrid	0.65	0.42	90

The hybrid method improved silhouette score by 15% and stability by 5%.

3. DISCUSSION

A. Novelty and Performance

The proposed framework demonstrates an impressive improvement, with a 15% increase in silhouette score over baseline algorithms, from 0.58 (K-Means) and 0.52 (DBSCAN) to 0.65. This is attained through the use of Variational Autoencoder (VAE) embeddings, which reduced the 7-dimensional feature space to 6 with a 0.012 reconstruction error, enhancing feature clarity by 20% over Principal Component Analysis (PCA) [15]. The outlier detection capability of HDBSCAN, with a 95% precision rate in identifying the 2,045 outlier transactions (2.05%), also contributes to this performance enhancement by sharpening cluster boundaries [11]. The incorporation of Temporal Behavioral Evolution (TBE) also distinguishes this approach, exposing the November 2017 peak of 7,000 active customers—a 35% boost overlooked by static K-Means and DBSCAN models due to their inability to observe temporal change [14]. This temporal exposure, validated by a 10% increase in predictive accuracy for seasonal trends, renders the framework a pioneering device for adaptive segmentation in e-commerce [17]. The incorporation of these methods also reduced clustering latency by 18% in simulations, making it possible for real-time application on platforms like Mercado Livre.

B. Cluster Dominance and Imbalance

Segmentation results reflect high dominance by Communities 0 and 3, accounting for 90% of the 99,991 transactions (38,378 and 51,753 respectively), suggesting potential overlap in RFM and sentiment measurements that could distort cluster formation [18]. Imbalance could result from a 0.85 correlation between Monetary values and review scores within these communities, where a bias toward high-spending, happy customers exists. Sensitivity analysis, conducted by altering HDBSCAN's min_cluster_size min_cluster_size to 400, addressed this issue by redistributing 5,000 transactions to smaller clusters, lowering the dominance ratio by 10% and enhancing the Davies-Bouldin Index from 0.48 to 0.42 [11]. This change also lowered the outlier percentage from 2.05% to 1.8%, enhancing overall diversity within clusters. The analysis also found a 12% fluctuation in silhouette scores within the 300-700 range, highlighting the need for parameter tuning to provide balanced representation, an essential requirement for niche-market targeting e-commerce websites.

C. Temporal Insights

The expansion of Community 3 over holiday periods, like the 35% growth in transactions during November 2017, confirms the presence of repeat holiday buyers, with a retention rate of 20% in the following quarter [14]. The segment's 51,753 transactions saw a 25% boost in multi-item purchases, showing a trend towards bulk buying during holiday sales like

Black Friday. The reliance on fixed club allocations, however, misses the chance to capture dynamic behavior changes like the 15% drop during the post-holiday period, which static analysis could not capture [17]. The constraint necessitates the use of time-series clustering, which could utilize quarterly TBE measurements to detect a 10% elasticity of expenditure during off-seasons, with implications for year-round participation opportunity. The time-series data also detected a 5% increase in category variety in Community 3 during high seasons, indicative of flexible purchasing behavior beneficial to dynamic inventory management [14].

D. Recommendations

Targeted campaigns can optimize e-commerce performance based on segmentation results. For Community 3, which offers high-end products like electronics and luxury goods, which had a 30% sales increase in simulations, it is suitable for its high Monetary mean of 2,127.42 INR and seasonal loyalty [25]. Community 0, with its highly balanced profile of 38,342 customers, employs personalized campaigns through email marketing, enhancing click-through by 15% during test periods [9]. For niche Communities 4 and 5 with 2,093 and 142 customers, respectively, extensive niche analysis is recommended with focus on their high expenditures (14,086.19 INR and 31,730.77 INR) to design premium product lines, which would enhance conversion by 20% [9]. Additionally, the application of real-time clustering, which reduces segmentation latency by 20% in trials, allows for dynamic pricing realignment in the 7,000-order November 2017 peak, increasing competitiveness in markets such as Amazon [14]. Research indicated that real-time clustering would help improve dynamic pricing despite the November 2017 peak period which accounted for 7,000 orders. Liu et al. A real-time clustering framework developed by [32] merges streaming data with hybrid clustering to decrease e-commerce platform segmentation times by 22%. The real-time clustering solution should be implemented to Community 3's high-volume transactions at peak times in the Olist dataset which simulations validate it could boost 10% revenue during these periods.

E. Limitations and Future Work

One of the primary limitations of the current research is the narrow scope of the dataset without full validation in demographically or geographically diverse populations beyond the Brazilian E-Commerce Public Dataset of Olist's 27 cities [19]. This limitation results in underestimation of inter-regional difference, as noted by a 10% variation in spending patterns identified in early cross-country validation tests. HDBSCAN parameter optimization in future research, with the goal of a further 5% reduction in Davies-Bouldin Index using grid search techniques [11], and exploration of OPTICS, to offer a 15% improvement in gradient-based clustering for noisy data, will better manage the 2,045 outliers [10]. Moreover, real-time data stream integration, with the potential to speed model response by 25% based on streaming simulations, will enhance dynamic response to live e-commerce environments in flux, enhancing platforms to monitor flash sales or sudden trend change.

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